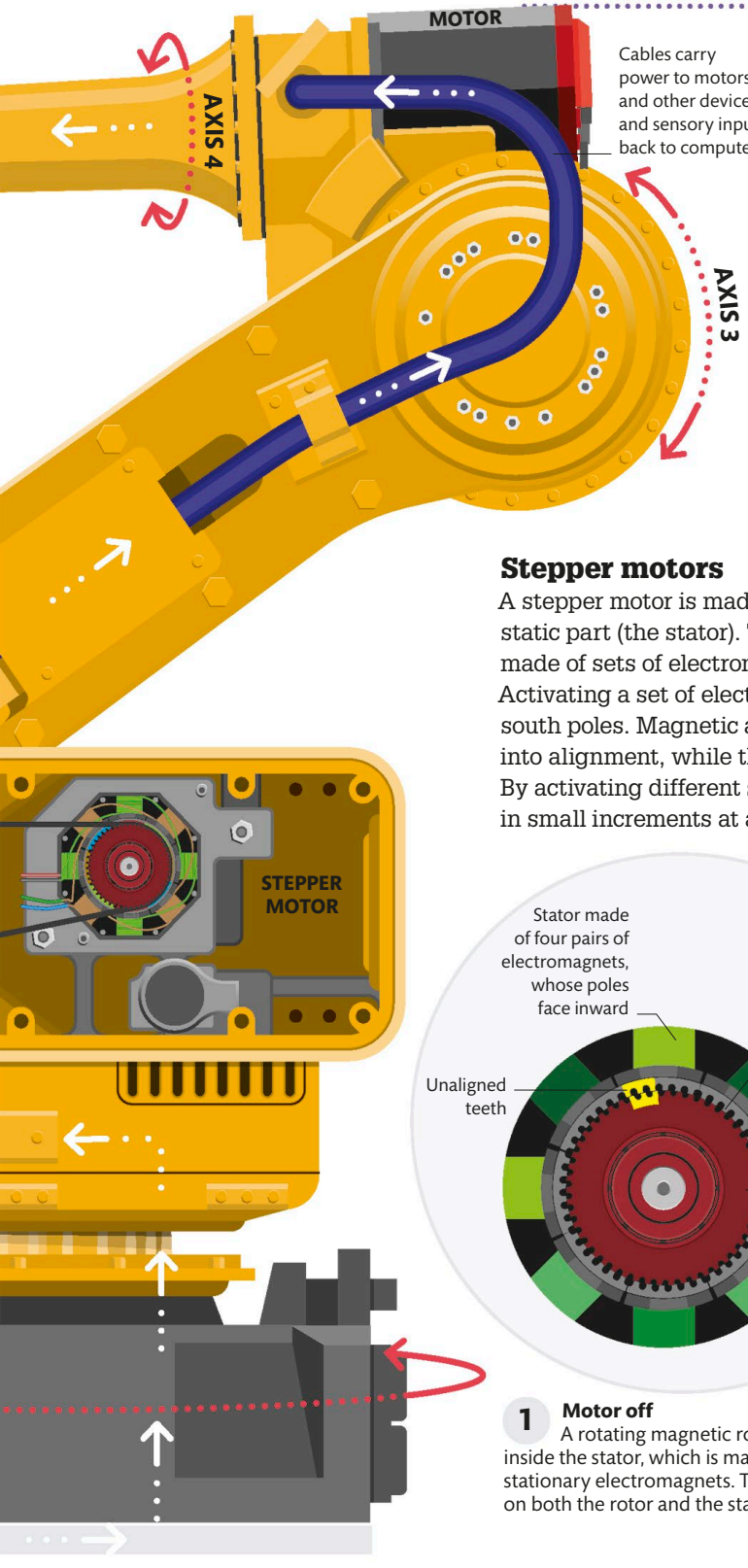
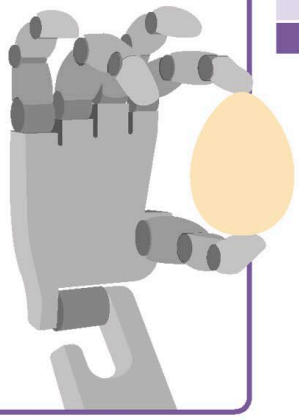


Cables carry power to motors and other devices, and sensory input back to computer

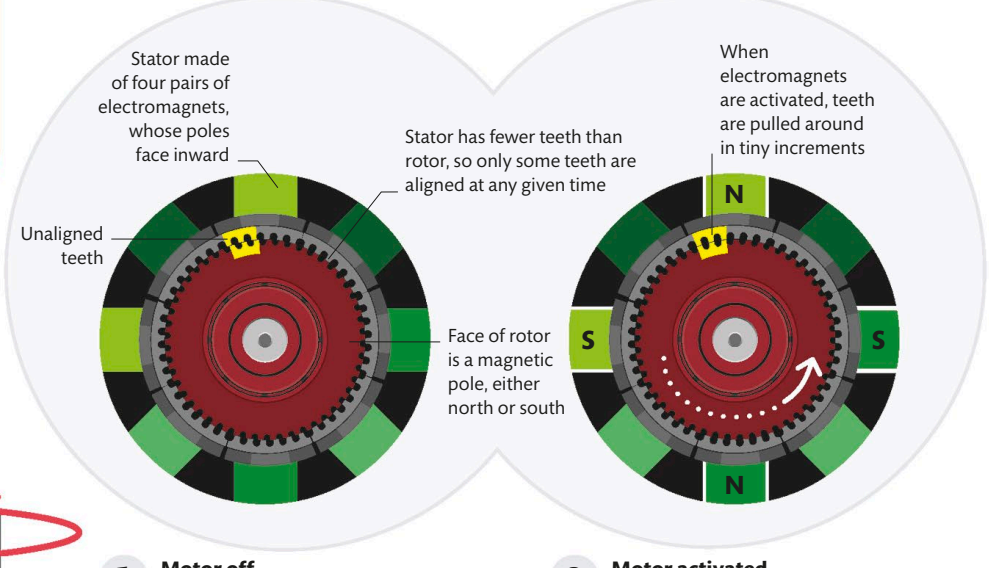
SENSING PRESSURE

The simplest type of pressure sensor used in robots is a foam pad that can conduct electricity, sandwiched between two metal plates. The plates are connected to a power supply. The more compressed the foam is, the more current flows through it.



Stepper motors

A stepper motor is made up of an inner rotating part (the rotor) and an outer static part (the stator). The rotor is a permanent magnet, and the stator is made of sets of electromagnets. The stator has fewer teeth than the rotor. Activating a set of electromagnets magnetizes the stator teeth with north and south poles. Magnetic attraction brings one set of teeth with opposing poles into alignment, while the matching poles are pushed out of alignment. By activating different sets of electromagnets, the rotor can be rotated in small increments at a time.



1 Motor off
A rotating magnetic rotor sits inside the stator, which is made of paired, stationary electromagnets. There are teeth on both the rotor and the stator.

2 Motor activated
When the electromagnets are activated, the magnetism drags the rotor around a tiny amount to make different teeth align. Each pair is activated in turn, moving the rotor.